

STAT 418 — FINAL PROJECT PROPOSAL

Airline Merger Damages Estimator

Estimating Consumer Harm from the AA / US Airways Merger using Causal Machine Learning

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Project Overview & Motivation

01

What This Project Does

- Replicates antitrust damages estimation work done by economic consulting firms in litigation.
- Quantifies consumer harm from the 2013 AA / US Airways merger using the but-for methodology standard in expert witness testimony.

Core Research Questions

- Did the merger raise ticket prices on routes where AA and US Airways previously competed?
- What is the total dollar value of consumer overcharges?
- Which routes and markets were harmed most?

Target Users

- Antitrust economists and policy researchers
- Airline market analysts evaluating domestic consolidation

Dec 2013

Merger closed

1,223

Routes where AA + US competed

622M

Affected passengers (2014–2017)

Data Collection Approach & Progress

02

Primary Data: DOT DB1B (BTS)

- Source: BTS Origin & Destination Survey
- Method: Programmatic download, no API key required
- 32 quarters (2010-2017), ~5M ticket records per quarter
- Final panel: 122,787 route-quarter observations

```
# Automated bulk download loop
url = f"transtats.bts.gov/PREZIP/"
      f"DB1BMarket_{year}_{quarter}.zip"
```

Control Variables: FRED API

- Jet fuel prices: Primary cost confounder
- Airline CPI: Macro price trend
- Free FRED API, single authenticated call per series

32

Quarters downloaded

4 GB

Raw data on disk

2

FRED series pulled

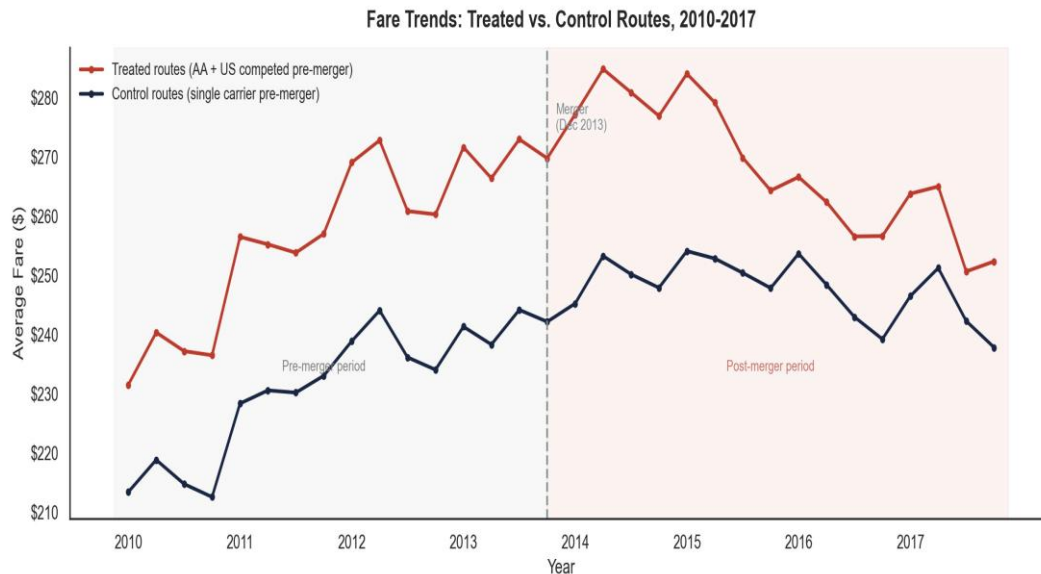
Sample DB1B Record

Route	Carrier	Fare (\$)	Pax
LAX-JFK	AA	452	41
PHL-CLT	US	189	28

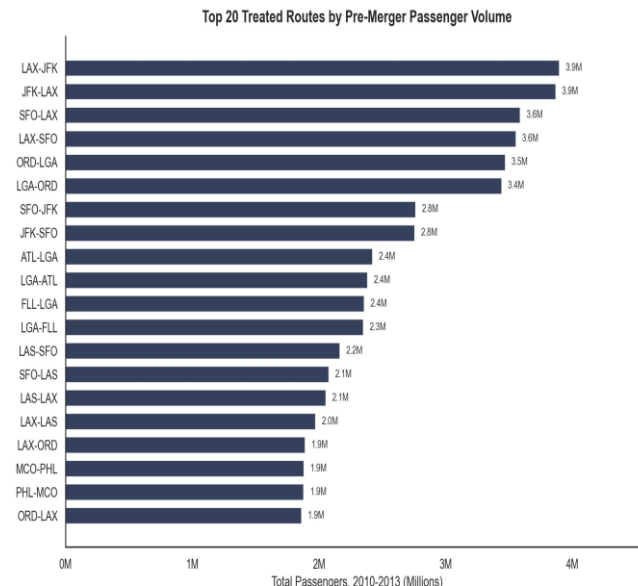
Exploratory Data Analysis

03

Fare Trends — Treated vs. Control, 2010–2017



Top 20 Treated Routes by Passenger Volume



1,223

Treated routes

3,721

Control routes

\$263

Mean treated fare

1,367mi

Mean treated dist

Proposed Modeling Approach

04



Outcome & Treatment

Outcome: Log average fare per route-quarter

Treatment: Indicator for routes where both AA + US operated pre-merger

Damages: (actual fare – counterfactual fare) x passengers = total consumer harm

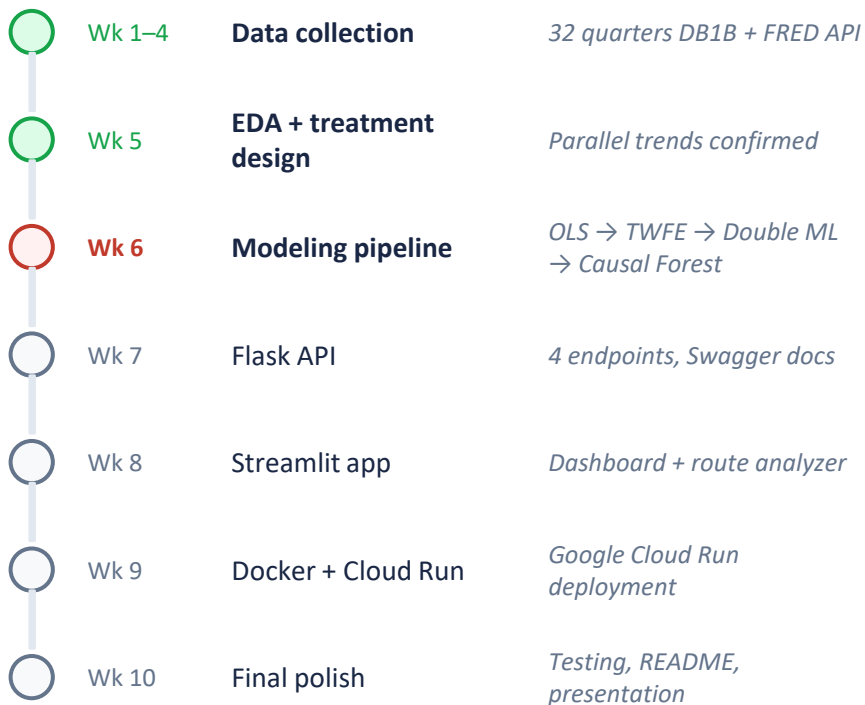
Key Controls (Double ML nuisance model)

- Jet fuel prices (log): Primary cost confounder
- Route distance: Structural demand driver
- HHI + carrier count: Competition controls
- Origin / destination state FE (encoded)
- Quarter time trend

Timeline & Solution Architecture

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Project Timeline



Risk: Double ML runtime — mitigated by validated 1-quarter sample confirming data quality before full run.

Solution Architecture

